

Multi-Agent LLM Evaluation of Counterfactual Explanations for Income Prediction

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Abstract

This project studies [selected AI method] in the context of [seminar topic]. We formulate the research question: [insert research question]. We review the relevant literature, implement the method, and evaluate it on [dataset/task]. Our results show that [main result], while also highlighting [main limitation].

Introduction and Problem Formulation

This project connects two seminar themes: **Counterfactuals** and **Syrtis**. It is related to the Counterfactuals seminar because we study counterfactual explanations for a binary income prediction model. Given a person whose income is predicted to be below \$50K, we use DiCE to generate counterfactual examples that would change the model prediction to above \$50K. The central question is not only whether these counterfactuals flip the model output, but whether they are good explanations. In the context of explainable AI, a useful counterfactual should be close to the original input, plausible, actionable, and sparse, meaning that it should change only a small number of features.

The project is also related to the Syrtis seminar because we implement a multi-agent LLM pipeline for evaluating these explanations. The seminar discussed generative AI workflows, orchestration of AI services, hallucination control, auditability, human validation, and trustworthy LLM systems. Our pipeline follows this perspective by using several LLM-based roles, including adversarial evaluators, a proof executor, a defense agent, and an expert witness, to assess counterfactual quality in a structured and auditable way.

The main research question addressed in this report is:

Do LLM-based evaluators, and in particular a multi-agent adversarial debate system, improve the assessment of counterfactual explanation quality compared with deterministic metrics?

This question is relevant because counterfactual generation methods such as DiCE typically select counterfactuals

using deterministic criteria. These include **validity**, which checks whether the generated counterfactual flips the model prediction; **proximity**, which favors small changes from the original input; **sparsity**, which favors changing few features; and **feasibility**, which can restrict which features are allowed to change. These metrics are necessary, but they do not fully capture whether a proposed change is realistic, fair, causally coherent, or convincing from a human perspective. We therefore investigate whether LLM-based evaluators can provide a richer assessment of the counterfactuals generated by DiCE.

We evaluate this question on the OpenML Adult Income dataset, where the task is to predict whether a person's income is above or below \$50K.

Related Work

Taxonomy of the Research Area

The research area of [topic] can be broadly divided into several families of methods:

- **Family 1:** [brief explanation];
- **Family 2:** [brief explanation];
- **Family 3:** [brief explanation].

This taxonomy helps position [selected method] within the broader literature.

Seminal and Recent Works

Compared with these methods, [selected method] is distinctive because [reason]. However, it also relies on assumptions such as [assumption], which may limit its applicability in [setting].

Method

We now describe [selected method]. The method aims to [goal of method]. At a high level, it works by [main algorithmic idea].

Given an input x , the method computes [output] by solving/estimating:

$$\hat{y} = f_{\theta}(x), \quad (1)$$

where f_{θ} denotes [model/method]. The key steps are:

Method	Metric 1	Metric 2	Metric 3
Baseline 1	–	–	–
Baseline 2	–	–	–
Selected method	–	–	–

Table 1: Experimental results comparing the selected method with baselines.

Step 1 ;
 Step 2 ;
 Step 3 .

In our implementation, we follow [paper/source] with the following adaptations: [describe simplifications, modifications, or implementation choices].

Experimental Setup

Dataset

We evaluate the method on [dataset name]. The dataset contains [brief description: number of samples, features, classes, task, etc.].

Baselines

We compare [selected method] against the following baselines:

- **Baseline 1:** [description];
- **Baseline 2:** [description].

Metrics

We use the following metrics:

- **Metric 1:** [why appropriate];
- **Metric 2:** [why appropriate];
- **Metric 3:** [why appropriate].

Implementation Details

The implementation is written in Python. The code is available at:

<https://github.com/your-username/your-repository>

The repository includes instructions to reproduce the experiments, including dependencies, dataset preparation, and commands used to generate the results.

Results and Analysis

Table 1 summarizes the main experimental results.

The results show that [main observation]. Compared with [baseline], the selected method [performs better/worse] in terms of [metric]. This suggests that [analysis].

However, the method also presents limitations. First, [limitation 1]. Second, [limitation 2]. These limitations are important because they affect [robustness, scalability, interpretability, feasibility, etc.].

Figure ?? illustrates [what the figure shows].

Discussion

The experiments indicate that [answer to research question]. The main strength of [selected method] is [strength]. This makes it suitable for [setting].

Nevertheless, the method assumes that [assumption]. In practice, this assumption may not hold when [case]. Another limitation is [another limitation].

Possible improvements include [future improvement], [extension], or combining the method with [other approach].

Conclusion

This report studied [selected method] for [task/problem]. We reviewed the relevant literature, implemented the method, and evaluated it experimentally. Our findings suggest that [main conclusion]. Future work could investigate [future direction].

References